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Purpose

In this work we examined the characteristic parameters of the optical fibre, in particular we investigated the operation of the two standard transmitter sources employed in optical fibre communication: the laser diode and the light emitting diode (LED).

* For all the samples provided in excel we calculated average and standard deviation.

8.1 Threshold current and bias point

First, we plotted the output power with both sources (led and laser) then we put in evidence the bias point and the threshold current. It is done to set the device throughout the experiment.

The bias current of the led corresponds to a drive current of 50mA. The bias current of the laser corresponds to the point where the output current is half the maximum. The threshold current, only observable in laser, corresponds to
the start of the zone of direct proportionality between the pumping rate and the power output.

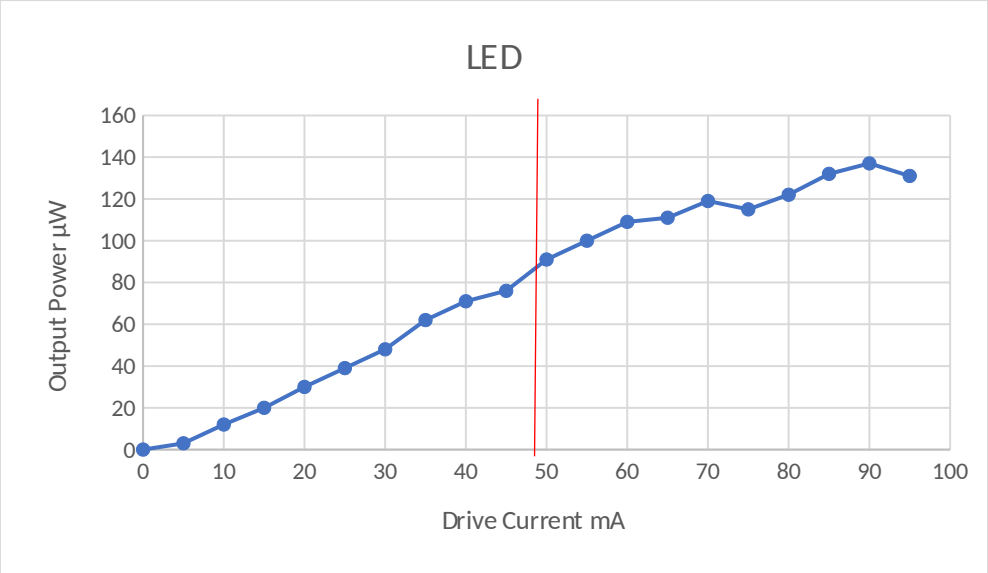


Figure 1 In evidence the bias point (50mA)

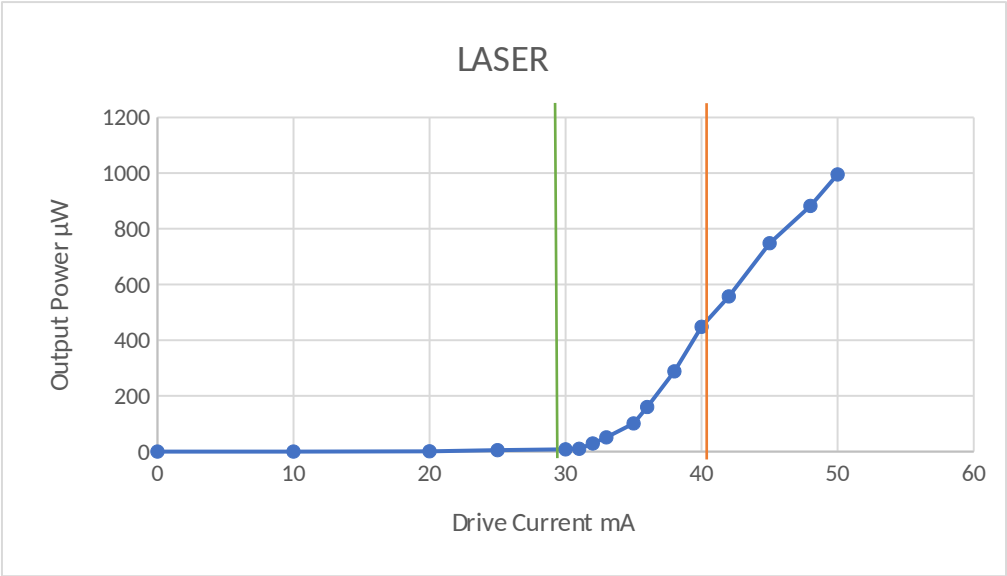


Figure 2 In evidence the threshold current (green) and the bias current (red)

8.2.2 Fibre attenuation coefficient

Knowing the launch power into the 1m patchcord for the led and the laser, we measured the total attenuation of the launched signal from each optical transmitter after propagation through fibre reel #1 and #2 for both the light sources.

Reel #1	dB	dB
Reel #2		

8.2.3 Length and attenuation per km of each optical fibre reel

In this part we estimated the length of the fibre knowing the delay between the signal directly from the waveform generator and the same signal transmitted via the optical fibre link.

Knowing the speed of light, the refractive index of the fibre and the time delay (time taken for the light to pass though the fibre) we calculated the length of the fibre for both reels.

By combining equations i) and ii) we obtain the fibre length as a function of the time delay.

Then we calculated the attenuation for km of the optical fibre using the results obtained in the previous point.

REEL #1		

REEL #2		
Average #1 #2		

8.2.4 Attenuation limited link lengths

Knowing the peak to peak amplitude of the launched optical signal (obtained by connecting the laser diode directly to the photoreceiver with the 1m pathcord), the dispersion coefficient and band amplitude of the noise signal, we calculated that is the max length that a path can have to get a useful signal to the receiver, along the way are also considered the losses.

= Peak to peak amplitude of the launched optical signal

= Band amplitude of the noise signal generated by the isolated photoreceiver. This value corresponds to the ratio between signal and noise power.

From literature we know that the sensitivity of a photodiode is the value of the power required to generate a SNR of 12.
So, we can say that must be at least 12 times the SNR.

REEL #1		
REEL #2		

These results represent the maximum length that still allows to get the minimum acceptable value of ~~the~~ power in the receiver.
For example, to have a minimum value of 252 mV using a led by reel #1, the fibre cannot exceed the length of about 2451 m, so we note that in this case a repeater should be placed before that length.

8.2.5

1.

Signal attenuation is caused by losses that occur during the transmission along the optic fiber. The losses are mainly due to Rayleigh scattering and absorption. The first phenomenon greatly reduces when λ increases (it is proportional to while absorption has a minimum around 1550nm. As result choosing a wavelength of 1550nm with the materials currently used corresponds to minimize the attenuation.

2.

State of the art fibers have attenuation coefficient of about 0.2db/km, or 0.0002dB/m, sensibly smaller than the coefficient we found in this project. Recalculating the attenuation limited link length in this situation knowing that:

We obtain

8.3

The data provided contain measurements relative to risetime of the transmitter/receiver and of the complete system considering the two fiber reels separately and combined. Measurements were obtained from the LED and the Laser.

To calculate and the following formulas were used:

1. Regarding the **LED** we obtain:

= 9,3 0,15ns		=9,080.38ns	=3,54ns
		=15,180.43ns	=5,92
		=24,060.60ns	=9,38

Using these results, we can calculate the bandwidth BW and the bit-rate BR.

The bandwidth is given by the equation:

The bandwidth length product is the product BW*L.

In a similar way the bit-rate can be easily calculated from the equation:

The Bandwidth-Length Product (BW.L) is obtained by:

The bit-rate distance (BR.L) is obtained by:

Knowing the length of reel 1# and reel#1 it's possible to calculate BW.L and BR.L

	BW	BW. L [MHz*m]	BR	BR.L [MHz*m]
#1	52,8MHz		70,6MHz	
#2	31,6MHz	14	42,2MHz	
#1+2	19,9MHz		26,6MHz	

2. Similarly for the **laser diode** we obtain

= 8 ns		= 0,37ns	=ns
		=ns	=4.8

	BW	BW.L [MHz*m]	BR	BR.L [MHz*m]
#1	64,7MHz		86,5MHz	
#1+2	40MHz		52MHz	

In the fibre the total rms impulse response is given by:

Where and are the rms impulse responses intermodal for material dispersion acting separately.

With laser we can consider the material dispersion much smaller than other one, in this case we can consider as total impulse response, the impulse intermodal response.

Finally, we can calculate the intermodal dispersion parameter for the laser as:

Vice versa, with LED the impulse response of material is significant, so we can consider as total impulse response, the impulse material response.

Finally, we can calculate the material dispersion parameter for the led as:

Where is the rms spectral linewidth of the led:

8.3.3

1)

In multimode fibres the principal dispersive mechanism responsible for pulse broadening are intermodal and material dispersion.

The intermodal dispersion only depends on the refractive indexes of the fibre. It does not depend on the wavelength or the source linewidth as it happens in the material dispersion. For the intermodal dispersion we must consider the *rms* widths and take it into account the distribution of energy among the modes.
If we consider a multimode step-index we can discard material dispersion since intermodal dispersion is much larger.
The laser has a total impulse response lower efficient than the led source, in fact the laser source is more efficient than the led one.
We can conclude that the laser has a bandwidth higher than the LED one because of a lower time delay of the light within the fibre.